☑ As a member of the University's student body, I will complete this assessment in a fair, honest, responsible and trustworthy manner. This means that:

- I will not seek out any unauthorised help in completing this assessment.
  (NB. Unauthorised help includes a tutorial or answer service whether in person or online, friends or family, etc.)
- I will not discuss or share the content of the assessment with anyone else in any form, including but not limited to, Canvas, Piazza, Chegg, Facebook, Twitter, Discord or any other social media within the assessment period.
- I will not reproduce and/or share the content of this assessment in any domain or in any form where it may be accessed by a third party.
- I am aware the University of Auckland may use Turnitin or any other plagiarism detecting methods to check my content.
- I declare that this assessment is my own work, except where acknowledged appropriately (e.g., use of referencing).
- I declare that this work has not been submitted for academic credit in another University of Auckland course, or elsewhere.
- I declare that I generated the calculations and data in this assessment independently, using only the tools and resources defined for use in this assessment.
- I declare that I composed the writing and/or translations in this assessment independently, using only the tools and resources defined for use in this assessment.

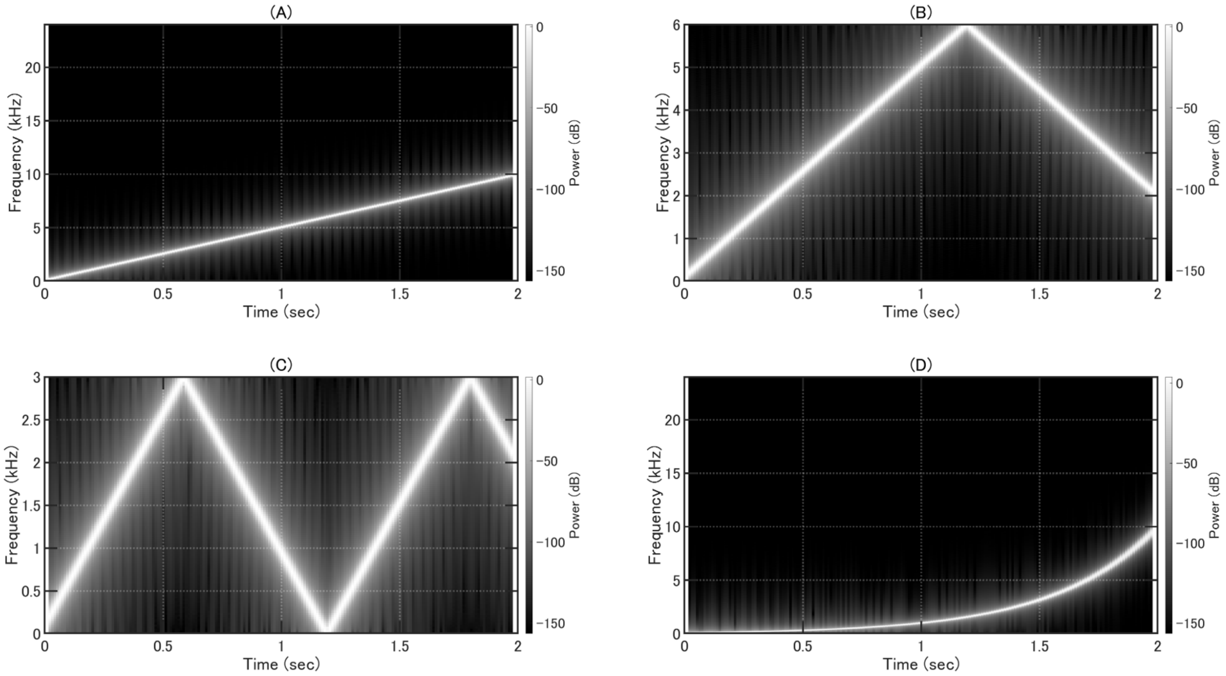
I understand the University expects all students to complete coursework with integrity and honesty. I promise to complete all online assessment with the same academic integrity standards and values.

Any identified form of poor academic practice or academic misconduct will be followed up and may result in disciplinary action.

Select one alternative

○ I agree

1 An acoustical engineer records the sound of various audio files played from a loudspeaker using a digital sound recorder equipped with a single microphone. Figures (A) to (D) below show the spectrograms of the recorded audio files. Answer following questions.



(i) Choose the correct spectrogram of the recorded sound when the linear swept sine signal (100 Hz to 10 kHz; 2 second length) is played from the loudspeaker and the sampling frequency of the recorder is set at 48 kHz.

(2 marks)

- (A)
- (B)
- (C)
- (D)

(ii) Choose the correct spectrogram of the recorded sound when the linear swept sine signal (100 Hz to 10 kHz; 2 second length) is played from the loudspeaker and the sampling frequency of the recorder is set at 6 kHz.

(3 marks)

- (A)
- (B)
- (C)
- (D)

(iii) When the linear swept sine signal (100 Hz to 10 kHz; 2 second length) is played from the loudspeaker, assume that the signal is recorded by pulse code modulation (PCM) with 24 bits/sample and the sampling frequency of the recorder is set at twice the Nyquist rate of the sound. Determine the size of the file saved on the recorder in bytes.
(3 marks)

[ ] bytes

Maximum marks: 8

2 Choose ALL metrics that represent the subjective loudness of sounds. **(2 marks)**

- [ ] Phon
- [ ] Time weighted sound level (no frequency weighting)
- [ ] Sone
- [ ] Mel

Maximum marks: 2

3 Assume that the musical notes across one octave are denoted as follows and the note "C" is tuned to f Hz.

C	C#	D	D#	E	F	F#	G	G#	A	A#	B

When equal temperament is used to define the pitches of these notes, which of the following frequencies the note "G" should be tuned to?
(2 marks)

- $7f$
- $\frac{7}{12}f$
- $7(2^{\frac{1}{12}})f$
- $2^{\frac{7}{12}}f$

Maximum marks: 2

4 Choose <u>ALL</u> processes that involve convolution. **(2 marks)**

- Calculating the frequency response of a microphone from its measured impulse response.
- Applying an FIR filter to remove noise from audio recordings.
- Simulating a recording of an arbitrary sound in a room using the impulse response between the positions of the sound source and microphone.
- Converting an analogue signal to a digital signal.

Maximum marks: 2

5 Choose <u>ALL</u> factors that affect speech intelligibility of a conversation that takes place in a room. **(4 marks)**

- [ ] Type of background noise in the room
- [ ] Listener's proficiency of the language spoken
- [ ] Spearker's pronunciation
- [ ] Amount of reverberation in the room
- [ ] Binaural hearing

Maximum marks: 4

6 The following statements are concerned about the physiology of human's auditory perception system. Answer if the statements are tue or false.

i) "The pinna is a part of the outer ear which helps us determine which angle a sound source is located."
(1 mark)

- ○ True
- ○ False

ii) "The role of the middle ear is to efficiently transmit the energy of sound from the ear drum to the cochlea."
(1 mark)

- ○ True
- ○ False

iii) "Outer hair cells convert the vibration of the basilar membrane to neural signals transmitted to the brain."
(1 mark)

- ○ True
- ○ False

iv) "The cochlea is the only organ in the inner ear."
(1 mark)

- ○ True
- ○ False

v) "*Cochlea amplifier* refers to an active function of the basilar membrane."
(1 mark)

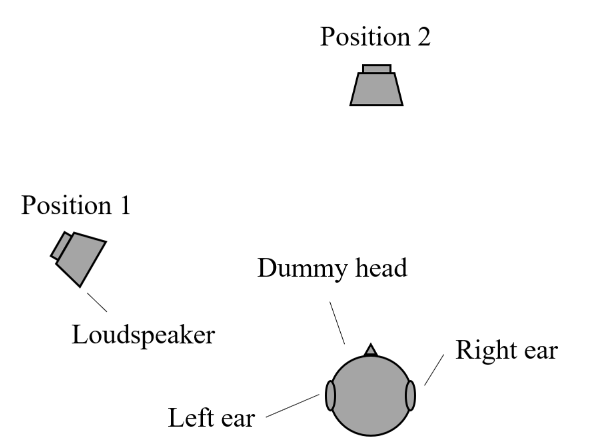
- True
- False

Maximum marks: 5

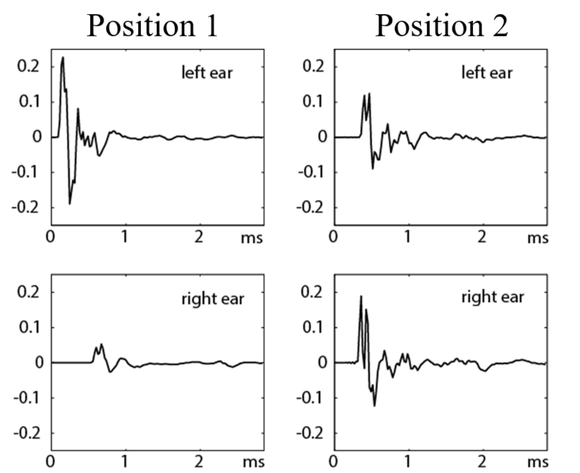
7 The following figure shows the head related impulse responses (HRIR) measured in an anechoic chamber (as in the plots on the right) by chainging the position of a loudspeaker while a dummy head was fixed (as in the diagram on the left).

List TWO binaural cues used in binaural hearing and explain how they relate to the shape of the HRIRs by referring to the positions of the loudspeakers, <u>in no more than 200 words</u>.

(6 marks)



Positions of the loudspeaker and dummy head



Measured head related impulse responses

Fill in your answer here

Format

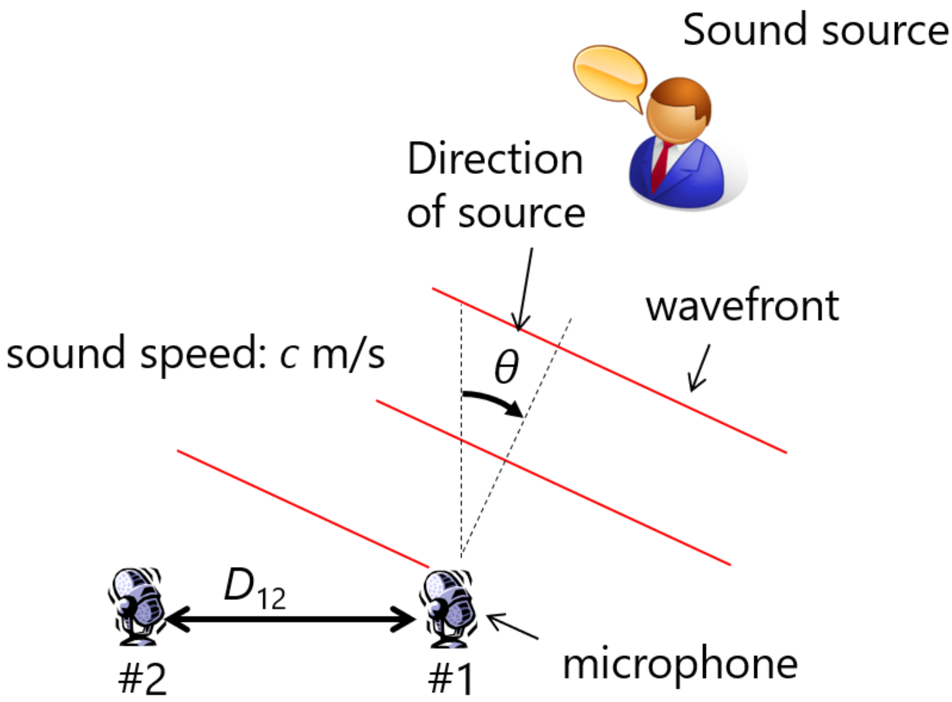
Words: 0

Maximum marks: 6

8 Time difference of arrival (TDOA) is the value used in microphone arrays. Answer the following questions.

i) Define TDOA by referring to the variables found in the following diagram, i.e. θ, c and D_{12}. **(1 mark)**

ii) Name ONE typical application of microphone arrays and briefly describe its mechanism by referring to how TDOA is utilised in the associated digital signal processing algorithms, <u>in no more than 150 words</u>. You may have equations/drawings in your answer (use formula/drawing functions in the answer panel) which are not counted towards the word limit. **(5 marks)**



Fill in your answer here. Clearly show the question number (i or ii) you are answering.

Format

Words: 0

Maximum marks: 6

An acoustical engineer calculated the reverberation time of a room between 125 Hz and 8 kHz using the T20 measure from a room impulse response measured in the room. The following description reports the procedures taken to derive the reverberation time.

“Firstly, the room impulse response was measured in the room using a loudspeaker and microphone. For the input signal (i.e. the sound played from the loudspeaker), a swept sine was used rather than a unit impulse because it is more robust to noise while **it satistfies the requirement**. Once the room impulse response was measured, a filterbank which consists of ___(i)___ filters was applied to the measured room impulse response. These digital filters ‘pass’ frequency components in every ___(ii)___, where the passband of the filter is centred either at 125, 250, 500, 1000, 2000, 4000 and 8000 Hz. A/An ___(iii)___ impulse response filter was used because it is less computationally expensive to achieve the required stopband drop-off of its frequency (amplitude) response. If the filtered room impulse response is denoted as $h(t)$, where t is the time in seconds, the early decay curve (EDC) was calculated by implementing the formula: ___(vii)___. The slope of the EDC was estimated using linear regression, which was applied to the time period between two points ___(iv)___ dB and ___(v)___ dB on the EDC below the initial level of the EDC. For example, the formula given by the linear regression at 1 kHz was $y = At + B$, where A = -80 dB/s, B = 20 dB and y is the linear fit to the EDC in dB. Hence, the reverberation time of the room at 1 kHz was determined as ___(vi)___ seconds.”

9 Fill the blanks (i) to (vi) with an appropriate word or value.

(i): **(1 mark)**

(ii): **(1 mark)**

(iii): **(1 mark)**

(iv): **(1 mark)**

(v): **(1 mark)**

(vi): **(1 mark)**

Choose the correct formula for the blank (vii). **(1 mark)**

- $\int_{-\infty}^{\infty} h(t)e^{-\omega t}dt$
- $\frac{\int_0^{50\mathrm{ms}} h^2(t)dt}{\int_{50\mathrm{ms}}^{\infty} h^2(t)dt}$
- $\frac{\int_0^{\infty} h^2(t)e^{-j\omega t}dt}{\int_0^{\infty} h^2(t)dt}$
- $\int_t^{\infty} h^2(\tau)d\tau$

Maximum marks: 7

10 State the requirement for the input signal used for measuring an impulse response (the boldfaced part in the description), in no more than 50 words. **(2 marks)**

Fill in your answer here

Format

Words: 0

Maximum marks: 2

11 The measured room impulse response also includes the characteristics of the loudspeaker and microphone used for the measurement. Summarise how one could remove the characteristics of the loudspeaker and microphone from the measurement, in no more than 150 words. Attatch a drawing of block diagram that explains how the characteristics of the measured room impulse response can be broken into. Assume the loudspeaker, microphone and room are all linear time-invariant systems.

(6 marks)

Put your answer on a blank sheet of paper. Scan and upload your file here. Maximum one file.

The following file types are allowed: **.pdf** Maximum file size is **1 GB**

Select file to upload

Maximum marks: 6

12 Choose ALL responses which are correct.

i) In an anechoic chamber - **(2 marks)**

- [ ] there is no sound
- [ ] occupants find it easy to localize where sound is coming from
- [ ] the wedges provide excellent insulation from outside
- [ ] there is negligible reverberation

ii) The decibel is used in acoustics because – **(3 marks)**

- [ ] it was invented by Alexander Graham Bell
- [ ] it compresses a large numerical range into more reasonable numbers
- [ ] it is in accordance with the Weber-Fechner Law
- [ ] it is 1/10 of a Bel
- [ ] it matches the logarithmic response of human hearing

iii) Steven's Law – **(2 marks)**

- [ ] uses the A-weighted sound pressure level in order to estimate loudness
- [ ] gives numbers which relate linearly to the loudness of subjective sounds
- [ ] expresses the loudness of sounds in phons
- [ ] is only applicable for tonal sounds

iv) The ear canal and the ossicles in the human ear – **(3 marks)**

- ☐ will contribute to noise induced hearing loss if we don’t give our ears sufficient rest
- ☐ provide us with directional information
- ☐ contribute to the impedance matching between air and the cochlea fluids
- ☐ increase the sensitivity of our hearing
- ☐ contribute to our Head Related Transfer Functions

v) Primary school children need reflecting surfaces nearer than do adults because – **(2 marks)**

- ☐ they are more distracted by noise from neighbouring children
- ☐ the hearing system of such children has yet to fully mature
- ☐ they are smaller
- ☐ their integration times are shorter than for adults

Maximum marks: 12

13 The following is an example of a Noise Limit from the Auckland Unitary Plan –

Noise arising from activities within Residential zones

Except where other more specific controls apply, the $L_{Aeq(15\,min)}$ noise level and maximum noise level (L_{AFmax}) arising from any activity in the residential zones measured at or within the boundary of an adjacent property site in the residential zones must not exceed the following levels limits:

Monday to Saturday	*7am-10pm*	*50dB $L_{Aeq(15\,min)}$*
Sunday	*9am-6pm*	*50dB $L_{Aeq(15\,min)}$*
All other times		*40dB $L_{Aeq(15\,min)}$*
		75dB L_{AFmax}

Explain what the subscripts, *Aeq(15 min)* and *AFmax*, mean and why there are two limits given for "All other times".

(5 marks)

Fill in your answer here

Format B I U x₂ x² Ω Σ

Words: 0

Maximum marks: 5

14 Given that the Resource Management Act distinguishes three severities of Noise – i.e. Reasonable, Unreasonable and Excessive what would be a logical basis for distinguishing them?

(3 marks)

Fill in your answer here

Format

Σ

Words: 0

Maximum marks: 3

15 Workers in a manufacturing industry are exposed to the following sound levels during their working day:

Time	Level (Leq)
8 – 10 am	94 dB(A)
10 – 10.30 am	62 dB(A)
10.30 – Noon	94 dB(A)
12 – 12.30 pm	64 dB(A)
12.30 – 2 pm	89 dB(A)
2 – 2.15 pm	62 dB(A)
2.15 – 4 pm	89 dB(A)

i) Determine whether it is legally acceptable for them to work without hearing protectors if you are given that the maximum Leq for 8 hours is set at 85 dB(A).

ii) What – in % - is the worker's noise dose?

(7 marks)

Fill in your answer here. Clearly show the question number (i or ii) you are answering.

Format

Σ

Words: 0

Maximum marks: 7

16 i) Show why the Intensity Level of an objective sound is numerically equal to the Sound Pressure Level but not to Sound Power Level.

ii) Determine the sound power Level of a sound source which is radiating omnidirectionally if at 20 m away outdoors it creates a Sound Pressure Level of 75 dB. What factors might potentially influence the accuracy of your answer?

iii) Why might it be much more difficult to predict the SPL at very close distances to the source?

(5 marks)

Fill in your answer here. Clearly show the question number (i or ii or iii) you are answering.

Format

Words: 0

Maximum marks: 5

17 If the equation for predicting the sound wave pressure radiated by a Simple Source is given by
$p = \frac{A}{r} e^{i(\omega t - kr)}$
explain
i) how are ω and k related.
ii) what A is and how to determine it from the sound power output of the source creating the wave.

(4 marks)
Fill in your answer here. Clearly show the question number (i or ii) you are answering.

Format B I U x_2 x^2 I_x Ω Σ

Words: 0

Maximum marks: 4

18 Draw a diagram showing how to determine if a wall surface or purpose-built reflector will provide a first-order, integrable reflection to a listener who is seated a distance of 20 metres from the speaker in a lecture theatre, and over what angle the surface will potentially direct such integrable reflections to listeners.

(2 marks)

Put your answer on a blank sheet of paper. Scan and upload your file here. Maximum one file.

The following file types are allowed: **.pdf** Maximum file size is **1 GB**

Select file to upload

Maximum marks: 2

19 A piccolo and a tuba are instruments designed for producing notes at different parts of the audible frequency range – the piccolo for very high frequencies and the tuba for very low frequencies. Explain

i) why they have different internal diameters.

ii) why they have different lengths.

(4 marks)

Fill in your answer here. Clearly show the question number (i or ii) you are answering.

Format

Words: 0

Maximum marks: 4

20 A wall having the acoustical data and dimensions given below separates two rooms each of volume 55 m^3 and reverberation times which have been carefully designed to have the same value of 0.6 s in all octave bands.

Using the General Insulation equation, determine the octave band spectrum of the reverberant sound in the receiving room if the other has a source in it which has the sound power level spectrum given below.

(8 marks)

Octave band (Hz)	125	250	500	1000	2000	4000
Source Sound Power Level (dB)	65	70	75	79	69	60
Wall TL (dB)	15	30	36	41	39	47

Wall data	Width 4 m	Height 3 m	Thickness 90 mm	Mass per unit area 40 kg

Fill in your answer here

Format B I U x_2 x^2 I_x Ω Σ

Words: 0

Maximum marks: 8